INCH-POUND

MIL-C-85049/7A
30 November 1990
SUPERSEDING
MIL-C-85049/7
16 January 1980
MS3189B
10 May 1982

MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, ENVIRONMENTAL, CABLE SEALING, 45°, CATEGORY 1A (FOR MIL-C-5015 CRIMP, MIL-C-26482 SERIES 2, MIL-C-81703 SERIES 3, AND MIL-C-83723 SERIES III CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-85049.

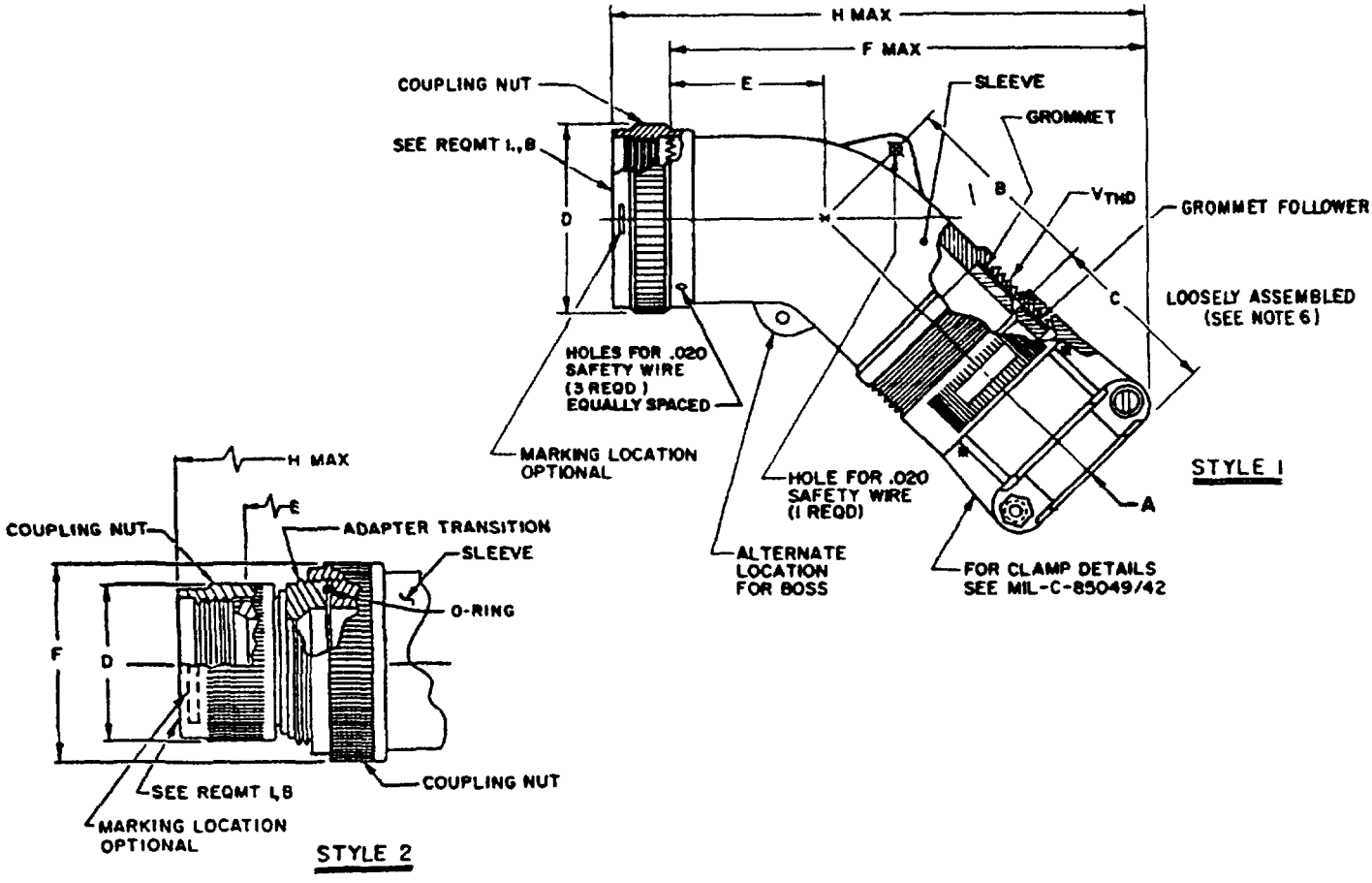


FIGURE 1. Configuration and dimensions.

AMSC N/A FSC 5935

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TABLE I. Dimensions.

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
1	3				4	.125	.250	1	1.790	1.027	.669
2					6	.250	.437	2	1.910	1.027	.669
3		8	8S	8	4	.125	.250	1	1.182	1.027	.617
4		10	10S, 10SL	10	4	.125	.312	1	1.850	1.027	.734
5					6	.250	.375	1	1.245	1.027	.734
6	7	12	12,12S	12	4	.125	.312	1	1.910	1.027	.858
7					6	.250	.437	1	1.910	1.027	.858
8					10	.350	.500	1	1.910	1.027	.858
9	12	14	14,14S	14	6	.250	.437	1	2.000	1.027	.984
10					10	.350	.575	1	1.655	1.027	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
1	.561	.625-24 UNEF	3.37	-	3.74
2	1.375	.750-20 UNEF	4.33	.858	4.70
3	.500	.625-24 UNEF	2.88	-	3.25
4	.561	.625-24 UNEF	3.41	-	3.78
5	.561	.750-20 UNEF	3.05	-	3.42
6	.625	.625-24 UNEF	3.52	-	3.89
7	.625	.750-20 UNEF	3.58	-	3.95
8	.625	1.000-20 UNEF	3.86	-	4.23
9	.688	.750-20 UNEF	3.71	-	4.08
10	.688	1.000-20 UNEF	3.74	-	4.11

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
11	19	16	16,16S	16	6	.250	.437	1	2.110	1.027	1.112
12					12	.500	.700	1	1.750	1.059	1.112
13	27	18	18	18	10	.350	.625	1	2.300	1.027	1.218
14					16	.625	.779	1	1.938	1.156	1.218
15	37	20	20	20	10	.350	.625	1	2.300	1.027	1.345
16					16	.625	.904	1	1.938	1.156	1.345
17		22	22	22	12	.500	.750	1	2.450	1.059	1.468
18					20	.875	1.029	1	2.093	1.375	1.468
19		24	24	24	12	.500	.750	1	2.450	1.059	1.593
20					20	.875	1.144	1	2.093	1.375	1.593

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
11	.781	.750-20 UNEF	3.88	-	4.25
12	.781	1.1875-18 UNEF	4.03	-	4.40
13	.968	1.000-20 UNEF	4.48	-	4.85
14	.968	1.4375-18 UNEF	4.49	-	4.86
15	.968	1.000-20 UNEF	4.48	-	4.85
16	.968	1.4375-18 UNEF	4.49	-	4.86
17	1.063	1.1875-18 UNEF	4.81	-	5.18
18	1.063	1.750-18 UNS	4.98	-	5.35
19	1.063	1.1875-18 UNEF	4.81	-	5.18
20	1.063	1.750-18 UNS	4.98	-	5.35

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
21			28		16	.625	.937	1	2.500	1.156	1.969
22					24	1.000	1.375	1	2.156	1.500	1.969
23			32		16	.625	.937	1	2.600	1.156	2.219
24					20	.875	1.250	1	2.600	1.375	2.219
25					28	1.250	1.625	1	2.250	1.781	2.219
26			36		16	.625	.937	1	2.690	1.156	2.469
27					24	1.000	1.375	1	2.690	1.500	2.469
28					32	1.437	1.840	1	2.344	1.830	2.469
29			40		16	.625	.937	1	2.690	1.156	2.719
30					24	1.000	1.375	1	2.690	1.500	2.719

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
21	1.188	1.4375-18 UNEF	5.11	-	5.64
22	1.188	2.000-18 UNS	5.09	-	5.62
23	1.250	1.4375-18 UNEF	5.24	-	5.77
24	1.250	1.750-18 UNS	5.52	-	6.05
25	1.250	2.250-16 UN	5.73	-	6.26
26	1.281	1.4375-18 UNEF	5.33	-	5.86
27	1.281	2.000-18 UNS	5.56	-	6.09
28	1.281	2.500-16 UN	5.95	-	6.48
29	2.031	1.4375-18 UNEF	6.08	-	6.61
30	2.031	2.000-18 UNS	6.31	-	6.84

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
31			40		32	1.437	1.875	1	2.344	1.830	2.719
32			44		16	.625	.937	1	2.690	1.156	2.969
33					24	1.000	1.375	1	2.690	1.500	2.969
34					32	1.437	1.875	1	2.344	1.830	2.969
35			48		16	.625	.937	1	2.690	1.156	3.219
36					24	1.000	1.375	1	2.690	1.500	3.219
37					32	1.437	1.875	1	2.344	1.830	3.219
38	61				12	.500	.750	1	2.450	1.059	1.653
39					20	.875	1.184	1	2.093	1.375	1.653
40	19	16	16,16S	16	10	.350	.625	1	2.110	1.027	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
31	2.031	2.500-16 UN	6.70	-	7.23
32	2.031	1.437-18 UNEF	6.08	-	6.61
33	2.031	2.000-18 UNS	6.31	-	6.84
34	2.031	2.500-16 UN	6.70	-	7.23
35	2.031	1.4375-18 UNEF	6.08	-	6.61
36	2.031	2.000-18 UNS	6.31	-	6.84
37	2.031	2.500-16 UN	6.70	-	7.23
38	1.063	1.1875-18 UNEF	4.81	-	5.18
39	1.063	1.750-18 UNS	4.98	-	5.35
40	.781	1.000-20 UNEF	4.16	-	4.53

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
41	27	18	18	18	4	.125	.312	1	2.300	1.027	1.218
42					6	.250	.437	1	2.300	1.027	1.218
43	37	20	20	20	6	.250	.437	1	2.300	1.027	1.345
44		22	22	22	4	.125	.312	1	2.450	1.027	1.468
45					6	.250	.437	1	2.450	1.027	1.468
46		24	24	24	10	.350	.625	1	2.450	1.027	1.593
47			36		12	.500	.750	1	2.690	1.059	2.469
48			40		12	.500	.750	1	2.690	1.059	2.719
49		10	10S, 10SL	10	6	.250	.437	2	1.910	1.027	.734
50	12	14	14,14S	14	12	.500	.750	2	2.300	1.059	.984

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
41	.968	.625-24 UNEF	4.13	-	4.50
42	.968	.750-20 UNEF	4.20	-	4.57
43	.968	.750-20 UNEF	4.20	-	4.57
44	1.063	.625-24 UNEF	4.34	-	4.71
45	1.063	.750-20 UNEF	4.40	-	4.77
46	1.063	1.000-20 UNEF	4.68	-	5.05
47	1.281	1.1875-18 UNEF	5.20	-	5.73
48	2.031	1.1875-18 UNEF	5.95	-	6.48
49	1.375	.750-20 UNEF	4.33	.858	4.70
50	1.717	1.1875-18 UNEF	5.36	1.218	5.73

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
51	19	16	16,16S	16	16	.625	.937	2	2.450	1.156	1.112
52	27	18	18	18	12	.500	.750	1	2.300	1.059	1.218
53	61				16	.625	.937	1	2.450	1.156	1.653
54	37	20	20	20	12	.500	.750	1	2.300	1.059	1.345
55					20	.875	1.250	2	2.500	1.375	1.345
56		22	22	22	10	.350	.625	1	2.450	1.027	1.468
57					16	.625	.937	1	2.450	1.156	1.468
58					20	.875	1.250	2	2.500	1.375	1.468
59		24	24	24	16	.625	.937	1	2.450	1.156	1.593
60			28		12	.500	.750	1	2.500	1.059	1.969

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
51	1.812	1.4375-18 UNEF	5.70	1.468	6.07
52	.968	1.1875-18 UNEF	4.61	-	4.98
53	1.063	1.4375-18 UNEF	4.95	-	5.32
54	.968	1.1875-18 UNEF	4.61	-	4.98
55	1.936	1.750-18 UNS	6.14	1.969	6.51
56	1.063	1.000-20 UNEF	4.68	-	5.05
57	1.063	1.4375-18 UNEF	4.95	-	5.32
58	1.936	1.750-18 UNS	6.14	1.969	6.51
59	1.063	1.4375-18 UNEF	4.95	--	5.32
60	1.188	1.1875-18 UNEF	4.97	--	5.50

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
61			28		20	.875	1.250	1	2.500	1.375	1.969
62			32		24	1.000	1.375	1	2.600	1.500	2.219
63			36		20	.875	1.250	1	2.690	1.375	2.469
64					28	1.250	1.625	1	2.690	1.781	2.469
65			40		20	.875	1.250	1	2.690	1.375	2.719
66					28	1.250	1.625	1	2.690	1.781	2.719
67			44		20	.875	1.250	1	2.690	1.375	2.969
68					28	1.250	1.625	1	2.690	1.781	2.969
69			48		20	.875	1.250	1	2.690	1.375	3.219
70					28	1.250	1.625	1	2.690	1.781	3.219

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
61	1.188	1.750-18 UNS	5.39	-	5.92
62	1.250	2.000-18 UNS	5.47	-	6.00
63	1.281	1.750-18 UNS	5.62	-	6.15
64	1.281	2.250-16 UN	6.08	-	6.61
65	2.031	1.750-18 UNS	6.37	-	6.90
66	2.031	2.250-16 UN	6.83	-	7.36
67	2.031	1.750-18 UNS	6.37	-	6.90
68	2.031	2.250-16 UN	6.83	-	7.36
69	2.031	1.750-18 UNS	6.37	-	6.90
70	2.031	2.250-16 UN	6.83	-	7.36

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TABLE I. Dimensions (continued).

Dash No.	Fits Shell of Connector				Clamp Size	A (see note 6) Wire Bundle Accommodation Range		Style	B ±.062	C Ref	D Dia Max
	MIL-C-81703 Ser 3	MIL-C-26482 Ser 2	MIL-C-5015 Crimp	MIL-C-83723 Ser III		Min	Max				
71	7	12	12,12S	12	10	.350	.625	2	2.110	1.027	.858
72	27	18	18	18	16	.625	.937	2	2.450	1.156	1.218
73		24	24	24	20	.875	1.250	2	2.500	1.375	1.593
74	12	14	14,14S	14	4	.125	.312	1	2.000	1.027	.984
75	19	16	16,16S	16	4	.125	.312	1	2.110	1.027	1.112

Dash No.	E ±.125	V Thread Class 2A	F Max	G Dia Max	H Max
71	1.529	1.000-20 UNEF	4.90	1.112	5.27
72	1.812	1.4375-18 UNEF	5.69	1.468	6.06
73	1.936	1.750-18 UNS	6.13	1.969	6.50
74	.688	.625-24 UNEF	3.64	-	4.01
75	.781	.625-24 UNEF	3.81	-	4.18

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INCH	mm	INCH	mm	INCH	mm	INCH	mm
0.005	0.13	1.375	34.93	3.410	86.61	5.180	131.57
0.020	0.51	1.437	36.50	3.420	86.86	5.200	132.08
0.062	1.57	1.4375	36.51	3.520	89.41	5.270	133.85
0.125	3.18	1.468	37.29	3.580	90.93	5.320	135.12
0.200	5.08	1.500	38.10	3.640	92.46	5.330	135.38
0.250	6.35	1.529	38.84	3.740	95.00	5.350	135.89
0.312	7.92	1.593	40.46	3.780	96.01	5.360	136.14
0.350	8.89	1.653	41.99	3.810	96.77	5.390	136.91
0.375	9.53	1.655	42.04	3.880	98.55	5.500	139.70
0.437	11.10	1.717	43.61	3.890	98.80	5.520	140.20
0.500	12.70	1.750	44.45	3.950	100.33	5.560	141.22
0.561	14.25	1.781	45.24	4.010	101.85	5.620	142.75
0.575	14.60	1.790	45.47	4.080	103.63	5.640	143.25
0.617	15.67	1.812	46.02	4.030	102.36	5.690	144.53
0.625	15.88	1.830	46.48	4.110	104.39	5.730	145.54
0.669	16.99	1.840	46.74	4.130	104.90	5.770	146.55
0.688	17.48	1.850	46.99	4.180	106.17	5.860	148.84
0.700	17.78	1.875	47.63	4.160	105.66	5.920	150.36
0.734	18.64	1.910	48.51	4.200	106.68	5.950	151.13
0.750	19.05	1.936	49.17	4.230	107.44	6.000	152.40
0.779	19.79	1.938	49.23	4.250	107.95	6.050	153.67
0.781	19.84	1.969	50.01	4.330	109.98	6.060	153.92
0.858	21.79	2.000	50.80	4.340	110.24	6.070	154.17
0.875	22.22	2.031	51.59	4.400	111.76	6.080	154.43
0.904	22.96	2.093	53.16	4.480	113.79	6.090	159.68
0.937	23.80	2.110	53.59	4.490	114.05	6.130	155.70
0.968	24.59	2.156	54.76	4.500	114.30	6.140	155.96
0.984	24.99	2.219	56.36	4.530	115.06	6.150	156.21
1.000	25.40	2.250	57.15	4.570	116.07	6.260	159.00
1.027	26.09	2.300	58.42	4.700	119.38	6.310	160.27
1.029	26.14	2.344	59.54	4.710	119.63	6.370	161.80
1.059	26.90	2.450	62.23	4.730	120.14	6.480	164.59
1.063	27.00	2.469	62.71	4.770	121.15	6.500	165.10
1.112	28.24	2.500	63.50	4.810	122.17	6.510	165.35
1.144	29.06	2.600	66.04	4.850	123.19	6.610	167.89
1.156	29.36	2.690	68.33	4.860	123.44	6.630	168.40
1.182	30.02	2.696	68.48	4.900	124.46	6.700	170.18
1.1875	30.16	2.719	69.06	4.970	126.24	6.830	173.48
1.188	30.18	2.880	73.15	4.980	126.49	6.840	173.73
1.218	30.94	2.969	75.41	5.030	127.76	6.900	175.26
1.245	31.62	3.050	77.47	5.050	128.27	7.230	183.64
1.250	31.75	3.219	81.76	5.090	129.29	7.360	186.94
1.281	32.54	3.250	82.55	5.110	129.79	7.390	187.70
1.345	34.16	3.370	85.60				

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REQUIREMENTS:

1. Design and construction:

   a. Dimensions and configuration: See figure 1 and table I.

   b. Interface dimensions shall conform to MIL-C-85049 figure 4.

   c. Figure 1 consists of a coupling nut and follower. The coupling nut shall be captivated to and free to rotate on the follower.

2. Material and finish:

   a. Accessory components shall be as shown in table II.

   b. Grommet and O-Ring - Grommet and O-ring elastomer material shall be in accordance with MIL-C-85049.

   c. Clamp assembly screws - Corrosion resistant steel in accordance with MIL-C-85049 passivated per QQ-P-35.

3. Qualification: See MIL-C-85049, category 1A. Information pertaining to qualification of products may be obtained from the Commanding Officer, Naval Avionics Center, Attention: Code B/444, 6000 East 21st Street, Indianapolis, IN 46219-2189.

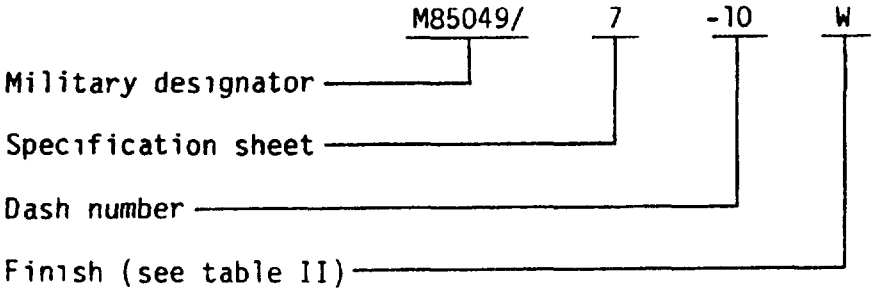
4. Applications:

   a. These accessories are for environmental applications.

   b. Style 2 applies when the cable diameter exceeds the "E" diameter of the interface dimensions defined in MIL-C-85049 figure 4.

   c. For assembly instructions see figure 2 of MIL-C-85049/11.

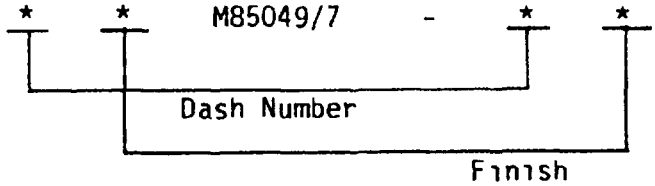
5. Part Number: See example below:



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6. Supersession data:

MS3189B * * M85049/7 - * *



Old	New
A	W
C	W

TABLE II. Material and finish.

Component	Material	Finish
Body	Aluminum alloy in accordance with MIL-C-85049	1/ N, 2/ W
Body	Corrosion resistant steel in accordance with MIL-C-85049	B, S

1/ Not for Navy use. Restricted to Air Force Space applications only.
2/ W finish is not for use in space application.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents shall be based upon 1.00 inch = 25.4 mm and shall be in accordance with ASTM E380, paragraph 4.5, Method A and ASTM E29, rounding-off method.
3. Unless otherwise specified, tolerances are ±.005 for three place decimals and ±.02 for two place decimals.
4. Dimensions apply after plating.
5. Loosely assembled is defined as the MIL-C-85049/42 clamp tightened to the MIL-C-85049/7 connector accessory body a minimum of two full thread lengths.
6. "Wire bundle accommodation range" is defined as the envelope area of the wire bundle. This dimension is not meant to define clamp hardware limits.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, MI
Navy - EC
Air Force - 11, 99
DLA - ES

Preparing activity:
Navy - AS
Project No. 5935-3729-04

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